1. Abstract

Recognizing everyday activities and sign-language gestures at home without cameras or body-worn devices would make smart environments both privacy-preserving and accessible to Deaf and hard-of-hearing residents. Existing WiFi Channel State Information (CSI) systems treat human activity recognition (HAR) and sign language recognition (SLR) as separate problems, and report near-perfect accuracy when the training and test data come from the same recordings. We present a single CSI pipeline that separates coarse daily activities from fine-grained sign gestures performed by one user. Two software-defined radios exchange IEEE 802.11 frames over one 20 MHz channel in the 2.4 GHz band, and the subcarrier amplitude of each received frame is segmented into short windows and normalized to suppress recording-specific bias. Two tasks are studied : motion versus no-motion using recurrence-plot images with a ResNet-18 and daily activity versus sign gesture using a lightweight one-dimensional convolutional network over 8 activities, 5 sign gestures and 2 baseline classes. Under subject-disjoint evaluation the first task reaches 96% accuracy, while under recording-disjoint evaluation the second reaches 71% accuracy and 0.71 macro-F1. The gap between the two shows that separating communicative gestures from ordinary motion, rather than detecting motion itself, is the harder problem on a single link.

2. Keywords

WiFi sensing, Channel State Information, Human Activity Recognition, Sign Language Recognition, device-Free sensing, Software-Defined Radio, convolutional neural networks

3. Introduction

Smart-home systems are being adopted quickly, with a projected 82.1% increase in installed systems by 2026 [CITE: Statista smart home market forecast]. At the same time, about 1.5 billion people live with hearing loss, a figure expected to reach 2.5 billion by 2050 [CITE: WHO World Report on Hearing]. For these residents, a home that responds to sign language rather than speech is not a convenience but a condition of access. The obvious way to read both daily activities and sign gestures is a camera, but 87% of studies in a recent scoping review reported privacy concerns with camera-based smart-home monitoring [CITE: JMIR scoping review on smart-home privacy], and cameras also need line of sight and adequate lighting. Body-worn sensors avoid the camera, but they must be worn, charged and remembered, which is a poor fit for continuous monitoring in a home [CITE: survey on wearable HAR burden].

WiFi Channel State Information (CSI) offers a third option. Every received WiFi frame carries an estimate of how the signal was attenuated and delayed on each subcarrier. When a person moves, the reflected paths change and these values vary over time. This variation is the sensing signal: it needs no camera, works in darkness, and records no image of the resident. Our link uses two Ettus USRP N210 software-defined radios (SDRs) running an IEEE 802.11 physical layer in GNU Radio on a single 20 MHz channel at 2.4 GHz, giving 52 subcarrier values per frame. The two radios share no common clock, so the phase is unusable and only amplitude is analysed.

WiFi CSI has been applied to both tasks, but separately. For activity recognition, deep models over CSI amplitude report above 99% inside the room they were trained in [CITE: ImgFi, 2023; PA-CSI, Quy et al. 2025], while domain-independent features raise cross-room accuracy to 83-92% at the cost of several synchronized links [CITE: Widar 3.0, Zheng et al. 2019]. For sign recognition, SignFi classified a large sign vocabulary at 98.9% for a known user, falling to 86.6% for unseen users [CITE: SignFi, Ma et al. 2018], while Wi-SL and RF-ASLR report 98.9% and 98.0%, each on a single signer in a controlled room [CITE: Wi-SL, Wang et al. 2023; RF-ASLR, Nosheen et al. 2026].

Three practical problems remain. First, a home that acts on sign language must know whether a movement was an ordinary activity or a deliberate message for communication, yet no reported WiFi system makes that decision: activity systems and sign systems are trained and tested apart. Second, the two kinds of movement differ in scale: a sign uses the hands and fingers, an activity moves the whole body, and one model must cover both. Third, those accuracies come either from testing on the same session used for training, or from receivers with several antennas placed around the room, so the same movement is measured from many angles at once. Our setup has one transmit and one receive antenna, so what a single pair achieves on an unseen person or recording is still unknown.

This paper studies that question using WiFi CSI as the only sensing modality, with one person in the room. One radio sends WiFi frames at a fixed rate and a second extracts the channel response from every frame; with it we collect two labelled datasets of daily activities and sign gestures. The amplitude is divided into short windows and normalized within each, so a model cannot recognize a recording instead of a movement. Two classifiers are then trained: one deciding whether motion is present, the other whether an observed movement is a daily activity or a sign gesture. Both are tested on people or recordings never seen during training, so the reported accuracy reflects unseen conditions. The work covers two objectives: separating activities from sign gestures with a single modality, and extending the same setup to complex, interleaved activities performed by one person .